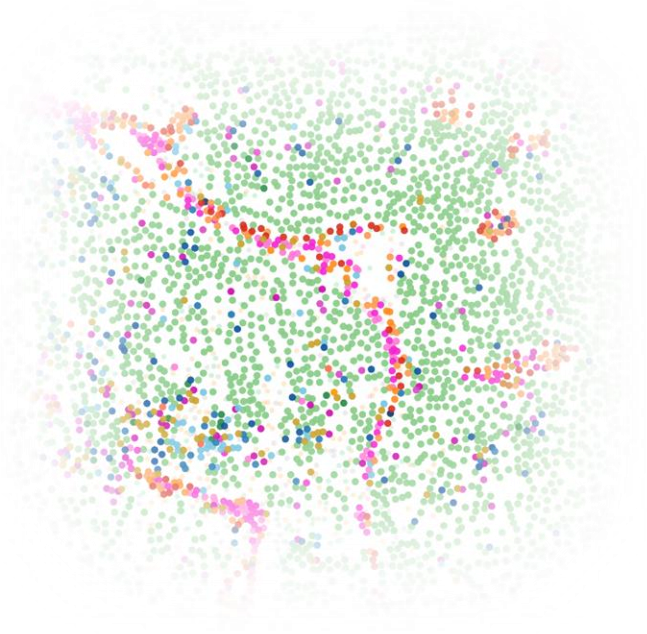
AARHUS UNIVERSITY

Benchmarking computational methods for spatial clustering in muscle-invasive bladder cancer CosMx spatial transcriptomics data



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I. Abstract

Background: The identification of cell-type populations and the characterization of their spatial organization within the tissue are fundamental to unraveling the role of the tumor microenvironment (TME) in cancer progression. In this context, spatial transcriptomics (ST) and spatial clustering methods emerge as powerful tools for generating tissue maps based on gene expression and cellular location. Multiple methods are available to perform spatially aware clustering of cell-type. Nevertheless, there is an important lack of benchmark studies that evaluate the suitability of these methods across different data types and consider the impact of the tissue architecture on their performance. Bladder cancer (BC) is a highly heterogeneous and prevalent disease, and the role of the TME is largely understudied. Evaluating and defining appropriate spatial clustering methods for ST data in BC could help establish methodological guidelines for future studies of the TME's composition and spatial organization.

Objectives: To evaluate the performance of BANKSY, BASS, and SEDR spatial clustering methods on muscle-invasive BC (MIBC) ST CosMx data. To assess the effect of method-specific parameter settings and data preprocessing steps on the performance. To define the set of parameters that improve the performance for each method and compare them. To interpret the clustering results generated in the context of the intrinsic spatial architecture of the tissue.

Methods: Tissue samples from 112 patients with muscle-invasive bladder cancer (MIBC) were processed as formalin-fixed paraffin-embedded tissue microarrays, yielding 298 tissue cores. Gene expression and spatial transcript locations were acquired using the CosMx Spatial Molecular Imager (Human 6K Discovery Panel). Cell segmentation was performed using membrane and nuclear protein markers. Data was preprocessed following Scanpy guidelines, and BANKSY, BASS, and SEDR were implemented according to official tutorials. Method performance was evaluated using: (i) alignment with reference cell type annotations (ARI, NMI, FMI, Homogeneity, Completeness, Purity); (ii) coherence with intrinsic spatial organization (PAS, CHAOS, ASW, SCS); (iii) robustness to data variation (sparsity, cellularity, number of clusters); and (iv) computational efficiency (runtime, memory usage).

Results: BANKSY exhibited a superior alignment with reference annotation and spatial architecture of the tissue compared to SEDR and BASS. BANSKY showed a balanced capture of both internal cluster composition and structural integrity. MIBC exhibits a granular spatial structure with low global spatial continuity, and the detection of finer clusters improved alignment with reference annotations. Data log-transformation influenced all methods' performance. BANKSY's performance was affected by λ parameter selection (best at $\lambda = 0.2$). All methods were robust to cellularity variation, and only BANSKY was impacted by gene expression matrix sparsity. BANKSY showed the best runtime scalability but the highest memory consumption.

Conclusions: BANKSY is the most suitable spatial clustering method for MIBC CosMx ST data among the methods evaluated. It effectively captures the intrinsic spatial architecture of the tissue, generating fine spatial clusters, which is reflected in an improved alignment with reference annotations. The predicted clusters overlap with identified niches in reference annotations, indicating the detection of cell state representations enriched with spatial information. The selection of λ parameter, the sparsity of the gene expression matrix, and data log-transformation impact the method's performance. Nevertheless, BANKSY is robust to cellularity variation and shows stable runtimes.

Keywords: tumor microenvironment, spatial transcriptomics, spatial clustering, bladder cancer, cell-type, benchmark.

VI. Methods

6.1. Dataset

6.1.1. Sample obtention

Serial sections of formalin-fixed paraffin-embedded (FFPE) tissue samples were obtained from 112 patients (females and males) with MIBC who underwent TURBT. All participants were included in the TOMBOLA trial, a Danish multicenter phase II clinical trial that investigates ctDNA guide adjuvant immunotherapy with atezolizumab after radical cystectomy (Dyrskjøt et al., 2026). The samples were prepared in eight TMAs that consisted of 35 – 47 ~1 mm diameter cores, with 13 – 17 patients and 1 – 3 cores per patient. The quality of the cores was assessed by an expert to ensure the inclusion of representative cores and non-disrupted tissue.

6.1.2. Dataset preparation

Following sample preparation, ST profiling was conducted using the CosMx Spatial Molecular Imager (NanoString) (He et al., 2022) and the Human 6K Discovery Panel. Single-cell segmentation was performed using CosMx platform, incorporating cell membrane and nuclear staining based on morphological protein markers. Different fields of view (FOVs) of interest were identified within the tissue. Unreliable negative probes, cells, and target genes were flagged according to the standard CosMx protocol. An Anndata object with 1.2 million cells per 6192 genes was generated. The dataset included (x, y) spatial coordinates and gene expression matrices for a total of 298 tissue cores.

6.1.3. Data preprocessing

Following standard Scanpy guidelines (Wolf et al., 2018), low-quality cells (less than 100 genes expressed) and lowly expressed genes (a minimum of 3 cells express the gene) were removed. The data was normalized (count depth normalization), and batch-aware feature selection based on TMA, retaining the top 3000 HVGs was performed. Log transformation and gene-wise scaling were only applied when it was suggested by the spatial method's official implementation tutorial.

6.2. Implementation of spatial clustering methods

We evaluated the performance BANKSY, BASS, and SEDR for spatial clustering of cell-types in CosMx MIBC data. These methods were selected based on the high performance, flexibility, and generalizability across tissue types and ST platforms reported in previous studies (R. Chen et al., 2025; Descoeudres et al., 2026; Hu et al., 2024; Yuan et al., 2024)

The methods were implemented by following the official tutorials (Li & Zhou, 2022; Singhal et al., 2024; H. Xu et al., 2024). Due to the large scale of the dataset, the methods returned to "out-of-memory" errors. Thereby, we ran each of the methods and assessed the performance of the clustering per tissue core.

Following the official guidelines for each method, we used the default settings for most parameters. Only the parameters described in this section were systematically evaluated to identify the optimal configuration of each method for our dataset (Table 5).

Table 5. Parameter settings used on BANKSY, BASS, and SEDR implementation.

Method	*Fixed clustering parameters*	*Fixed data-related parameters*	*Tested parameters*
BANKSY	• Number of neighbors: 15 • Clustering resolution: 0.4 • Partitioning graph algorithm: Leiden • nbr_weight_decay: scaled_gaussian	• Feature selection: 3000 HVGs with Scanpy • Scaling with z-score gene-wise	• Data preprocessing: log-transformation/non-transformation of normalized data • PCs retained: 20, 50 • Lambda value: 0.2, 0.4, 0.6, 0.8
BASS	• Number of neighbors: 15 • Expected number of cell-type labels: 28 • Partitioning graph algorithm: K-means • Number of MCMC iterations for refinement: 10 • Number of MCMC burn-in iterations: 2000	• Feature selection: 3000 HVGs with Scanpy	• Data preprocessing: log-transformation/non-transformation of normalized data • PCs retained: 20, 50 • Number of spatial domains (R): 3, 6
SEDR	• Number of neighbors: 6 • Expected number of clusters: 28 • Partitioning graph algorithm: K-means	• Feature selection: 3000 HVGs with Scanpy • Scaling with z-score gene-wise	• Data preprocessing: log-transformation/non-transformation of normalized data • PCs retained: 50, 200

6.3. Benchmark evaluation metrics

The evaluation of the method's performance included (i) an evaluation based on the alignment with reference manual annotations; (ii) coherence with the expected spatial continuity of our MIBC tissue types. To conduct these analyses, widely used evaluation metrics were computed to assess the method's performance.

In this section, we provide a detailed description of the metrics used. To facilitate its understanding, we divide the evaluation metrics into two groups: accuracy metrics (used to evaluate the alignment with reference annotations) and spatial metrics (which evaluate the spatial continuity and homogeneity of the clusters).

6.3.1. Accuracy metrics

ARI (Adjusted Rand Index) evaluates the pairwise similarity between the predicted clusters and the expected clusters (based on reference annotations). It assesses whether a pair of cells that belonged to the same cluster in reference annotations is also in the same cluster in the prediction (true positives or TP), and whether pairs of cells belonging to different clusters in reference annotations are correctly separated in the prediction (true negatives or TN). The misalignment of pairs is represented by the number of false positives (FP) and false negatives (FN). ARI corrects for random chance and ranges from -1 to 1, where 1 represents perfect agreement, 0 indicates random clustering, and -1 suggests clustering worse than random. The expected value of the index (E) indicates the value obtained in the case of random clustering.

$$ARI = \frac{TP + TN - E}{TP + TN + FP + FN - E},$$

$$E = \frac{(TP + FP) \times (TP + FN) + (FN + TN) \times (FP + TN)}{TP + TN + FP + FN}.$$

NMI (Normalized Mutual Information) measures the agreement of the predicted clusters with the expected clusters based on the amount of information that is shared between them, normalizing for different cluster sizes. This measurement is based on Mutual Information (MI), which assesses how much knowledge of one of the variables lowers the uncertainty of the other, based on entropy. NMI ranges from 0 to 1, where 0 indicates random clustering and 1 perfect agreement.

$$NMI = \frac{MI(A,B)}{\sqrt{H(A)H(B)}},$$

$$MI(A,B) = \sum_{i}^{|A|}\sum_{j}^{|B|} P(A_i, B_j)\log\left(\frac{P(A_i, B_j)}{P(A_i)P(B_j)}\right),$$

where $P(A_i)$ represents the probabilities of each cluster in the reference annotations (A) and $P(B_j)$ in the predicted grouping (B); A_i is the cluster i in the reference annotations, and B_j the cluster j in the predicted clustering; $P(A_i, B_j)$ the joint probability, $H(A)$ and $H(B)$ the entropies of A and B.

FMI (Fowlkes-Mallows Index) determines the similarity between predicted and expected clustering based on precision (positive predictive rate, PPR) and recall (true positive rate, TPR). Specifically, it assesses how concordant is the clustering predicted between pairs of cells related to the clustering of the reference annotations. FMI ranges from 0 to 1, where 1 indicates a high degree of similarity between clusters and 0 suggests different cluster assignments.

$$FMI = \sqrt{PPV \times TPR} = \sqrt{\frac{TP}{TP + FP} \times \frac{TP}{TP + FN}}$$

Homogeneity evaluates the internal consistency of the predicted clusters compared to clusters defined by the reference used. Homogeneity ranges from 0 to 1, where 1 indicates that all cells within a predicted cluster share the same cell-type label from reference annotations, and values closer to 0 suggest that predicted clusters show a heterogeneous composition of cell-types according to reference annotations.

$$Homogeneity = 1 - \frac{H(A|B)}{H(A)},$$

$$H(A|B) = -\sum_{j}^{|B|} P(B_j) \sum_{i}^{|A|} P(A_i|B_j) log P(A_i|B_j),$$

where *H(A)* is the entropy of the reference clustering, *H(A|B)* measures the uncertainty of the reference clustering after knowing the predicted grouping based on conditional entropy.

Completeness measures the cohesion of the predicted clusters relative to reference clustering. Completeness ranges from 0 to 1. A clustering result reaches a higher completeness, i.e. values closer to 1, when all cells that share cell-type labels in the reference annotations are grouped into the same cluster. In contrast, values closer to 0 may suggest over-fragmentation of the clusters based on reference annotations.

$$Completeness = 1 - \frac{H(B|A)}{H(B)},$$

$$H(B|A) = -\sum_{i}^{|A|} P(A_i) \sum_{j}^{|B|} P(B_j|A_i) log P(B_j|A_i),$$

where *H(B)* is the entropy of the predicted clustering, *H(B|A)* measures the uncertainty of the predicted clustering after knowing the reference clustering based on conditional entropy.

Purity evaluated the agreement between the predicted clustering results and the reference clustering by counting the proportion of the predominant cell-type label in the reference among the cells in each predicted cluster. Purity ranges from 0 to 1, where 1 represents a perfect agreement, and 0 indicates an unrelated cluster.

$$Purity = \frac{1}{n}\sum_{j} max_i \left|A_i \cap B_j\right|,$$

where A_i represents the cluster *i* in the reference annotations, B_j is the *j* cluster in the predicted clustering results, and *n* is the number of cells.

Average accuracy rank summarizes the accuracy achieved by a predicted clustering across all evaluated accuracy metrics, giving each metric equal weight. It is computed as the mean of the ranks obtained for each metric, where a lower score indicates better overall agreement with the reference annotations.

$$Average\ accuracy\ rank = \frac{1}{6}\left(Rank_{ARI} + Rank_{NMI} + Rank_{FMI} + Rank_{Homogeneity} + Rank_{Completeness} + Rank_{Purity}\right).$$

6.3.2. Spatial metrics

PAS (Proportion of Abnormal Spots) measures the homogeneity and spatial dispersion of the identified clusters based on the local cell neighborhood. It is computed by determining the proportion of cells whose assigned label is different from at least six to ten nearest neighbors. PAS ranges from 0 to 1, where lower scores indicate less spatial dispersion or more homogeneous clustering.

$$PAS = \frac{n_{abnormal}}{n}$$

where $n_{abnormal}$ represents the number of cells with abnormal labels and *n* the total number of cells evaluated.

CHAOS (Cluster Homogeneity And Spatial Organization Score) evaluates how compact or dispersed the cells of a single cluster are within the physical space. The metric is computed as the mean distance of the 1-Nearest Neighbor (1NN) graph constructed between cells sharing the same label for every identified cluster. CHAOS ranges from 0 to N/A. Lower CHAOS scores indicate a high compactness of the clusters identified.

$$CHAOS = \frac{1}{N}\sum_{k=1}^{K}\sum_{ij}^{n_k} w_{kij}\,,$$

$$w_{kij} = \begin{cases} d_{kij}, \text{if spot } i \text{ and } j \text{ are connected} \\ \text{in the 1NN graph in cluster } k, \\ 0, \quad \text{otherwise} \end{cases}$$

where d_{kij} represents the Euclidean distances between the cells *i* and *j* in the cluster *k; N* is the total number of cells evaluated, and *K* the total number of identified spatial clusters; and w_{kij} is the distance between connected cells in the 1NN graph for cluster *k*.

ASW (Average Silhouette Width) assesses the global spatial continuity of the clustering based on the geometric separation of the clusters within the physical space. It evaluates how well-defined and distinct the clusters are by comparing the similarity of cells within a cluster to cells in the neighboring clusters. ASW scores range from -1 to 1, where higher values indicate that clusters are well defined and clearly separated from each other in the space (high spatial continuity).

$$ASW = \frac{\sum_i^N SW_i}{N},$$

$$SW = \frac{b - a}{\max(a, b)},$$

where *SW* is the Silhouette Width determined for each cell; *a is* the mean distance between a cell and all other cells belonging to the same cluster; *b is* the mean distance between a cell and all other cells in the nearest cluster; *i* is the cell evaluated; and *N is* the total number of cells.

SCS (Spatial Coherence Score) measures the frequency with which adjacent cells share the same cluster label. It is based on O'Neill's spatial entropy, which quantifies the randomness of labels in the space. Higher SCS scores indicate that adjacent cells frequently have identical labels, reflecting a higher global spatial continuity.

$$H(G,L) = -\sum_{a,b\in K} P(\{a.b|E)\log\left(P(\{a,b\}|E)\right),$$

$$G = (V,E),$$

where *H(G,L)* is the spatial entropy calculated for a specific graph *G* and a given set of labels *L; V* is the set of all cells, and *E* is the set of edges that connect every pair of adjacent cells in the space; $P(\{a.b|E)$ is the joint probability that two adjacent cells share labels *a* and *b;* and a is the cluster label of one cell, and *b is* the cluster label of the neighboring cell.

6.4. Reference annotations

Manual in-house cell-type annotations were obtained for the analyzed dataset. These annotations were generated by performing naïve Leiden clustering, and cell-types were identified based on the differential expression of widely used genes for immune, epithelial, and stromal cell-types, as well as protein markers. Different rounds of annotations were performed to identify the different cell-subtypes and refine the annotations. The biological coherence of the clusters' granularity was assessed based on Silhouette Score (SS) and Davies-Bouldin Index (DBI) metrics. Both metrics evaluate the quality of the clustering algorithm based on the similarity of each cluster to other clusters based on intra-cluster and inter-cluster distances. The number of clusters that obtained higher SS and DBI scores (indicating more compact, well-separated clusters) was retained.

Ultimately, 28 cell-type labels were identified (Figure 31), which can be broadly group into cancer cells ("LumP", "LumNS", "B/Sq", "LumU", "NE-like"), T-cells ("CD-4 memory", "CD4-helper", "CD-8 cytotoxic", "CD-4 cycling"), other immune types ("TREM2/SPP1_TAM", "Myeloid_other", "Macrophage_core", "Monocyte_CCR2/FCN1+", "Plasma", "Neutrophil_like", "Immune_low", "B_cell"), fibroblasts ("mCAF", "CAF_APC", "myCAF", "CAF_APC_like", "Stroma_rich", "Stroma_low"), epithelial cells ("PLVAP+_TEC", "Inflamatory_TEC", "Cycling_TEC", "APC", "epithelial_low") and ambiguous ("low_signal").

Additional niche annotations were generated by an expert using scNiche algorithm (Dr. FAN et al., 2024), which uses a deep learning framework to identify niches based on cell's own transcriptome, neighborhood transcriptome, and neighborhood composition.

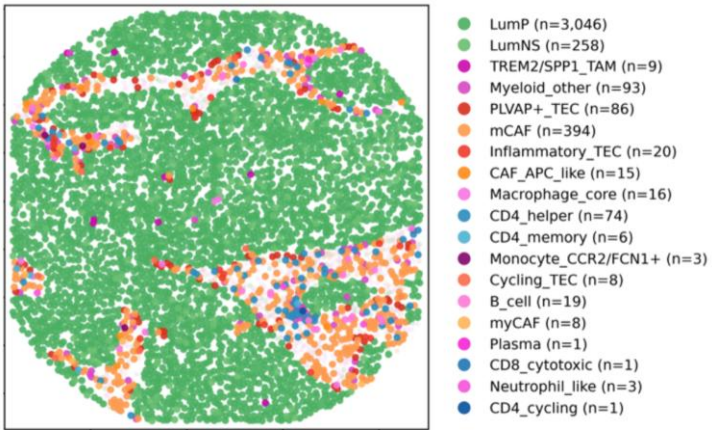


Figure 31. Representative tissue core labeled using reference cell-type annotations (courtesy of Jakob Kohler).

6.5. Software and libraries

Scanpy libraries (version 1.11.5) (Wolf et al., 2018)were used to analyze and preprocess the dataset. Python (version 3.11.13) was used for data analysis and figure generation. The *spicy* package (Sperotto et al., 2024) was used to calculate Pearson correlation coefficients, and *Scikit-learn* (Pedregosa et al., 2018)was used to compute all the described evaluation metrics. For Python-based methods (BANKSY and SEDR), *time.time()* was used to measure the runtime by recording the wall-clock time before and after the methods' execution. The memory usage was captured using *tracemalloc.get_traced_memory()*. In case of BASS, which is implemented on R, *Sys.time()* was used to measure the runtime, and *gc()* to capture memory usage.

VIII. Conclusions

In the present study, the performance of BANSKY, BASS, and SEDR was evaluated in the context of MIBC CosMx ST data. This evaluation involved the alignment with reference annotations (using ARI, FMI, NMI, Homogeneity, Purity, and Completeness metric scores), coherence with intrinsic tissue architecture (based on PAS, CHAOS, ASW and SCS metric scores), robustness to data variation (sparsity of gene expression matrices, cellularity, and number of clusters), and computational efficiency (memory usage and run time per tissue core). In contrast with previous studies, we conduct a data-specific benchmark and interpret the predicted clustering results in the context of their coherence with the underlying spatial tissue organization, providing insight into each method's predictions. Ultimately, we offer guidance on the spatial clustering method selection for our data.

According to the proposed aims and results obtained, we address the following conclusions:

- MIBC tissue samples show a global low spatial continuity, reflected by a granular structure where no uniform homogeneous spatial blocks can be identified.
- BANKSY is more appropriate for the analysis of our MIBC ST CosMx data than BASS and SEDR.
    - BANKSY showed an improved alignment with cell-type reference annotations based on accuracy scores.
    - BANKSY exhibited a more balanced performance profile compared to BASS and SEDR. Significantly strong correlations were detected between Purity (Pearson $r = 0.754$, $p < 0.001$), Homogeneity ($r = 0.929$, $p < 0.001$), and Completeness ($r = 0.708$, $p < 0.001$), which suggests that BANSKY prioritized both capturing the internal composition of the clusters and their structural integrity.
    - BASS and SEDR exhibited moderate correlations between ARI and Homogeneity (BASS: $r = 0.257$, $p < 0.001$; SEDR: $r = 0.490$, $p < 0.001$), which may indicate that the prediction of more complete clusters (that group all cells that share cell-type labels) is prioritized over capturing the internal composition of the clusters.
    - BANSKY consistently outperformed SEDR and BASS in terms of coherence with the intrinsic spatial architecture of the tissue, based on the distribution of spatial metric scores.
- BANKSY identifies fine spatial clusters, enriching cell state representations with spatial information. This was reflected by an improved alignment of the predicted clusters with those identified on the reference annotations using scNiche tool.

- The number of principal components retained did not have a considerable effect on SEDR, BASS, and BANSKY performance in our data.
- Data log-transformation did influence the performance of BANSKY, BASS, and SEDR, being more notable in the latter.
- Method-specific parameter setting may impact the predicted clustering result.
    - The selection of parameter λ on BANKSY did significantly affect the method's performance in terms of alignment with reference annotations and coherence with intrinsic spatial continuity. Among the tested values (λ = 0.2, 0.4, 0.6, and 0.8), the best results were obtained with λ = 0.2.
    - The choice of R parameter value in BASS (R = 3 or 6) had minimal impact on the methods' performance.
- The number of cells per sample was not associated with ARI scores but showed significant correlations with CHAOS and SCS metrics, suggesting an influence on cluster compactness and spatial organization.
- The prediction of more and finer clusters generally led to improved alignment with reference annotations according to ARI and Completeness scores.
- The sparsity of the gene expression matrix was correlated to a worse agreement with cell-type reference annotations only for BANKSY.
- The runtime and memory usage per tissue core increased with the number of cells for all methods.
    - BANKSY demonstrated the best scalability in terms of runtime; however, it also exhibited the highest memory consumption across most cellularity ranges.